RAJ GOLAKIYA

GitHub | LinkedIn

TECHNICAL SKILLS

- **Programming Languages**: Python, JavaScript
- **Machine Learning Frameworks**: Scikit-Learn, TensorFlow, Keras, PyTorch
- **Data Manipulation and Analysis**: NumPy, Pandas, SciPy
- **Natural Language Processing**: NLTK, SpaCy, Transformers, Beautiful Soup
- **Web Development**: HTML, CSS, Flask, ReactJS, NodeJS, ExpressJS
- **Databases**: SQL, MongoDB, PostgreSQL
- **Development Tools**: Git, GitHub, Jupyter Notebooks, Docker, Goole Colab, Kaggle
- **Additional Skills**: Data Analytics, Data Visualization (Tableau, PowerBI, Matplotlib, Seaborn), Supervised & Unsupervised Learning Algorithms, EDA, Feature Engineering, Feature Selection, Class Balancing, Feature Extraction, Algebra, Probability, Statistics, Calculus, Matrices

EDUCATION

Artificial Intelligence and Machine Learning
Lambton College Jan ’23 – Aug ‘24

- Investigated data science, Python programming, big data basics, AI career pathways, professional communication, and applied AI principles to practical problems.
- Examined data visualization techniques and tools, advanced Python for AI/ML, data mining concepts, Natural Language Processing fundamentals, and effective management of AI/ML projects.
- Studied neural network basics and practical applications, explored social media analytics, learned essential software tools for AI/ML, and completed a collaborative AI/ML capstone project with real-world applications and competitions.

Bachelor of Computer Application
Veer Narmad South Gujarat University Aug ’18 – Jun ’21

- Established foundational skills in communication, mathematics, and computer basics during the initial semesters.
- Developed proficiency in diverse programming languages such as C, Java, PHP, ASP.Net, and undertook advanced projects, including a thesis where developed a website using the MERN stack (MongoDB, Express.js, React.js, Node.js).
- Explored advanced topics like computer graphics, network technology, e-commerce, and cyber security, enhancing practical application capabilities.
- Culminated with a comprehensive understanding of computer applications, incorporating software engineering, system management, and hands-on project development with technologies spanning UNIX, Shell Programming, PHP, MySQL, ASP .Net, and the MERN stack.

Professional Experience

- Developed a full-featured e-commerce platform with an inbuilt wallet system for seamless user transactions, wallet top-ups, and purchase payments using Node.js, Express, and MongoDB**.**
- Designed and implemented a returns and refunds management system, automating refund processing to wallet/original payment method with proper order tracking and status management.
- Built a shipping integration system to manage order shipments, tracking, shipping APIs, and delivery status updates using REST APIs and third-party logistics integrations**.**
- Implemented a platform advertising system allowing sellers/admin to create promotional banners, featured products, and advertisement placements to increase product visibility.
- Developed secure authentication and authorization using JWT and role-based access control for admin, sellers, and customers.
- Designed responsive frontend using React.js / Next.js (or mention your frontend tech) and backend REST APIs with Node.js, ensuring scalable and modular architecture.

PROJECTS

IntelliExam: Intelligent Exam Paper Generator
AI/ML Project Developer

- **Overview**: Developed an innovative web-based tool to automate the generation of exam papers, leveraging advanced Natural Language Processing (NLP) and machine learning models.

- **User Input**: Designed a user-friendly interface where users can either provide a topic name or input their own content. Options include selecting the number of questions (10, 20, or 30) and the complexity level (Easy, Medium, Hard).

- **Data Processing**: Implemented robust backend processes using Flask, where user inputs are validated and processed. For topic-based inputs, integrated the Wikipedia API to fetch relevant content, using BeautifulSoup for parsing and spacy for enhanced topic modeling.

- **Model Development**: Evaluated multiple models, ultimately selecting GPT-3.5 from OpenAI for its superior performance in generating high-quality, customizable questions. Previous models like LLaMA3 and custom-built models were tested but found inadequate due to biases, limited capabilities, and high time complexity.

- **Question Generation**: Utilized the GPT-3.5 model to generate questions, specifying the desired number and complexity. Questions were formatted into a JSON response, then converted into a downloadable Microsoft Word document using python-docx.

- **Challenges and Solutions**: Overcame challenges in generating related topics and gathering comprehensive text corpora by implementing a combination of API calls and NLP techniques. Addressed model limitations by iterating on different solutions and integrating user feedback for continuous improvement.

- **Technologies Used**: Python 3.9, Flask 2.1.3, BeautifulSoup 4.10.0, NLTK 3.6.7, SpaCy 3.2.3, OpenAI API (GPT-3.5), pandas 1.3.5, requests 2.27.1, transformers 4.22.2, Docker, AWS EC2, Heroku.

- **Outcome**: Successfully created a scalable, efficient, and user-friendly tool that saves educators significant time and effort while providing high-quality, customizable exam papers. Received positive user feedback and identified areas for future enhancements.

Data Science Project
Lambton College

- **Objective**: Collaborated with data scientists and analysts to gather, clean, and preprocess large datasets for analysis and modeling as part of an academic project.

- **Exploratory Data Analysis (EDA)**: Conducted EDA to uncover insights, patterns, trends, and anomalies in the data.

- **Machine Learning Models**: Designed and implemented machine learning models such as linear regression, decision trees, and random forests to predict customer behavior and enhance marketing campaigns.

- **Data Visualization**: Created data visualizations using tools like Tableau and PowerBI to communicate findings and insights to stakeholders, aiding their comprehension of complex datasets.

- **Collaboration and Documentation**: Engaged in regular team meetings to discuss project progress, address challenges, and brainstorm potential solutions, promoting a collaborative and innovative work atmosphere. Documented project methodologies, data preprocessing steps, and model outcomes to ensure reproducibility and facilitate knowledge sharing within the team.